

Matthew C. Brennan

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Education

	Harvard University
2022	Ph.D. in Earth & Planetary Sciences
2020	M.A. in Earth & Planetary Sciences
	University of Chicago
2017	B.S. with Honors in Geophysical Sciences / B.S. in Environmental Sciences

Research Positions

2024 – now	Scientist Static High Pressure Team, Los Alamos National Laboratory
2022 – 2024	Harold Agnew National Security Postdoctoral Fellow Static High Pressure Team, Los Alamos National Laboratory
2017 – 2022	Graduate Research Assistant Laboratory for Mineral Physics, Harvard University
2016 – now	Synchrotron X-ray user HPCAT & GSECARS, Advanced Photon Source, Argonne National Laboratory Beamline 12.2.2, Advanced Light Source, Berkeley National Laboratory
2015 – 2017	Laboratory Technician Laboratory for Mineral Physics, University of Chicago
2016	Department of Energy SULI Researcher Energy Systems Division, Argonne National Laboratory

Publications

2026	“Crystal Structure, Compressibility, and Thermal Expansion of Uranium Mononitride” <u>M. C. Brennan</u> , M. A. Mathis, E. Kardoulaki, B. L. Musico, A. J. Nichols, S. W. Paisner, R. Ripani, B. T. Sturtevant, D. A. Rehn (2026). <i>Physical Review B</i> 113, 134116.
2025	“Crystal Structure and Thermal Expansion of Ta₂O₅ from Neutron Diffraction” <u>M. C. Brennan</u> , S. C. Vogel, B. T. Sturtevant (2025). <i>Physical Review Materials</i> 9, 073605.
2025	“Nonlinearity of the Post-Spinel Transition and its Expression in Slabs and Plumes Worldwide” J. Dong, R. A. Fischer, L. P. Stixrude, <u>M. C. Brennan</u> , K. Daviau, T. Suer, K. M. Turner, Y. Meng, V. B. Prakapenka (2025). <i>Nature Communications</i> 16, 1039.

- 2024 **“Phase Comparison and Equation of State for Ta₂O₅”** M. C. Brennan, D. A. Rehn, L. Q. Huston, B. T. Sturtevant (2024). *Journal of Physics: Condensed Matter* 36, 275401.
- 2023 **“Thermal Equation of State of U₆Fe from Experiments and Calculations”** M. C. Brennan, J. D. Coe, S. C. Hernandez, S. Crockett, L. Q. Huston, S. M. Thomas, B. T. Sturtevant, E. D. Bauer (2023). *Physical Review B* 108, 064108.
- 2023 **“Comparisons of the Core and Mantle Compositions of Earth Analogs from Different Terrestrial Planet Formation Scenarios”** J. Gu, R. A. Fischer, M. C. Brennan, M. Clement, S. A. Jacobsen, N. A. Kaib, D. P. O’Brien, S. N. Raymond (2023). *Icarus* 394, 115425.
- 2022 **“Water Storage Capacity of the Martian Mantle Through Time”** J. Dong, R. A. Fischer, L. Stixrude, C. Lithgow-Bertelloni, Z. T. Eriksen, M. C. Brennan (2022) *Icarus* 385, 115113.
- 2022 **“Timing of Martian Core Formation from Models of Hf–W Evolution Coupled with N-body Simulations”** M. C. Brennan, R. A. Fischer, F. Nimmo, D. P. O’Brien (2022) *Geochimica et Cosmochimica Acta* 316, 295–308.
- 2021 **“High-Pressure Deformation of Iron–Nickel–Silicon Alloys and Implications for Earth’s Inner Core”** M.C. Brennan, R. A. Fischer, S. Couper., L. Miyagi, D. Antonangeli, G. Morard (2021). *Journal of Geophysical Research: Solid Earth* 126, e2020JB021077.
- 2020 **“Equation of State of TiN at High Pressures and Temperatures: A Possible Host for Nitrogen in Planetary Mantles”** K. Daviau, R. A. Fischer, M. C. Brennan, J. Dong, T. Suer, S. Couper, Y. Meng, V. B. Prakapenka, (2020). *Journal of Geophysical Research: Solid Earth* 126, e2020JB020074.
- 2020 **“Core Formation and Geophysical Properties of Mars”** M. C. Brennan, R. A. Fischer, J. C. Irving (2020). *Earth and Planetary Science Letters* 530, 115923.

Presentations

- 2025 **“Recent High-Pressure Experiments that Shed Light on Planetary Cores”**, APS SCCM Meeting (Invited)
- 2024 **“Synthesis and Characterization of Metastable Tetragonal UTe₂ at Ambient Conditions”**, Gordon Research Conference
- 2024 **“High-Pressure Investigations of U-bearing Intermetallics”**, Energy and Natural Resources Security Group, Los Alamos National Laboratory (Invited)
- 2023 **“Thermal Equation of State of U₆Fe”**, APS SCCM Meeting
- 2023 **“The First Thermal Equation of State for U₆Fe”**, Dynamic Material Properties Meeting, Los Alamos National Laboratory

- 2023 **“Update on the U₆Fe Equation of State”**, Production Science Chemistry L2 Review, Department of Energy (Invited)
- 2022 **“High-Pressure Deformation of Iron–Nickel–Silicon Alloys and Implications for Earth’s Inner Core”**, Materials at Extreme Conditions Group, Stony Brook University (Invited)
- 2022 **“Static Deformation of Iron–Nickel–Silicon Alloys at High Pressures”**, Shock & Detonation Physics Group, Los Alamos National Laboratory (Invited)
- 2022 **“A Mineral Physics Perspective on the Martian Core”**, Planetary Geophysics Group, ETH Zürich (Invited)
- 2021 **“A Mechanically Strong Inner Core Implied by Deformation of Silicon-bearing Alloys”**, AGU Fall Meeting
- 2021 **“Deep Mars”**, Harvard EPS Colloquium (Invited)
- 2020 **“High-Pressure Deformation and Texturing of Fe–Ni–Si alloys”**, COMPRES Annual Meeting
- 2019 **“Martian Core Formation: Implications from the Hf–W System”**, Goldschmidt Conference
- 2019 **“Using Core Formation and Geophysical Modelling to Predict the Core Radius and Seismic Properties of Mars”**, Lunar and Planetary Science Conference
- 2018 **“A Core Formation Model with Implications for the Properties of the Martian Interior”**, AGU Fall Meeting
- 2018 **“The Composition and Seismic Properties of the Martian Interior”**, Goldschmidt Conference
- 2017 **“Deep-Earth Partitioning between Molten Iron Alloys and Solid Silicates”**, AGU Fall Meeting

Teaching & Mentoring

- 2026 – now **Co-mentor for Postdoctoral Research Associate** (LANL)
- 2024 – 2025 **Mentor for Graduate Research Assistant** (LANL)
- 2021 **Head Teaching Fellow**, A Brief History of the Earth (Harvard EPS 10)
- 2021 **Teaching Fellow**, Stellar and Planetary Astronomy (Harvard ASTRON 16)
- 2020 **Teaching Fellow**, A Brief History of the Earth (Harvard EPS 10)
- 2019 **Teaching Fellow**, Mineralogy (Harvard EPS 142)

Honors & Awards

2024	APS Student and Dissertation Award (American Physical Society GCCM)
2023	Harold Agnew National Security Postdoctoral Fellowship (LANL)
2022	Glenn T. Seaborg Nuclear Science Fellowship (LANL)
2021	Derek Bok Center Teaching Certificate (Harvard University)
2019	National Science Foundation Graduate Research Fellowship
2017	Departmental Honors in Geophysical Sciences (University of Chicago)
2014 – 2017	Dean's List (University of Chicago)

Service & Outreach

2025 – now	Executive Committee Member (HPCAT, Advanced Photon Source)
2025	Co-chair (APS GCCM Early Career and Student Symposium)
2025	Site Review Committee Member (NNSA SSAA Centers of Excellence)
2025 – 2027	Executive Committee Member (American Physical Society GCCM)
2024 – now	Student Research Symposium Judge (LANL)
2023 – 2025	Users' Executive Committee Member (Advanced Photon Source)
2023 – now	Proposal reviewer (NSF Chemical Evolution of the Solid Earth, ANR Appel à Projets Générique, NASA Emerging Worlds, Advanced Photon Source User Meeting, NNSA SSAA Centers of Excellence)
2022	Curatorial assistant (Harvard Mineralogical & Geological Museum)
2020 – now	Journal reviewer (<i>Science</i> , <i>APL Materials</i> , <i>Journal of Applied Physics</i> , <i>Planetary Science Journal</i> , <i>Physics Letters</i> , <i>Physics & Chemistry of Minerals</i> , <i>Nature Reviews: Earth & Environment</i> , <i>Nature Communications</i> , <i>Physics of the Earth & Planetary Interiors</i>)
2020	Science Education Partner (Harvard Museum of Natural History)
2018 – 2019	Graduate Outreach Chair (Harvard Earth & Planetary Sciences)
2017 – 2022	Laboratory Safety Officer , Department of Environmental Health & Safety (Harvard University)